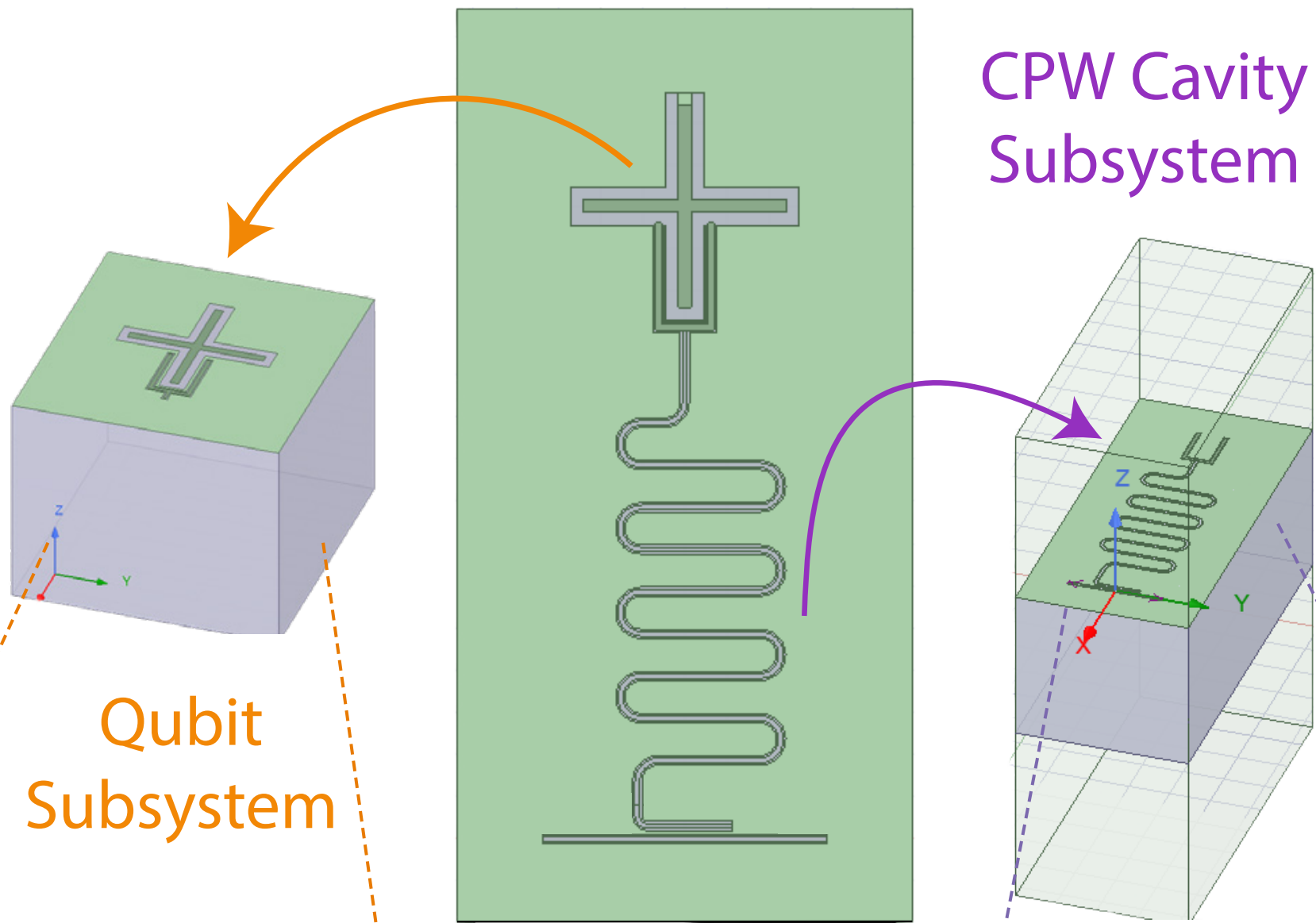




CPW Cavity  
Subsystem

Qubit  
Subsystem

$\alpha$  (anharmonicity)

$\approx -E_C$  from Q3D

$f_{\text{qubit}}$  (qubit frequency)

$\approx \sqrt{(8 E_J E_C) - E_C}$  from Q3D

$f_{\text{res}}$  (resonator freq.)

HFSS eigenmode solver

$\kappa$  (resonator linewidth)

HFSS via 50 $\Omega$  port loss